Spiked Wigner model

Consider $X = \lambda vv^T + W$:

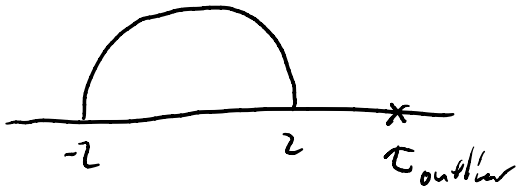
- W is Wigner
- $\|v\|_2 = 1$ and $\lambda > 0$.

What is the behavior of eigenvalues/eigenvectors of X?

By eigenvalue interlacing:

$$\lambda_n(X) \quad \lambda_{n-1}(X) \quad \lambda_2(X) \quad \lambda_1(X)$$
$$\lambda_n(W) \leq \lambda_{n-1}(W) \leq \cdots \leq \lambda_1(W) \leq$$

Recall: $\lambda_1(W) \to 2$ and $\lambda_n(W) \to -2$ a.s. as $n \to \infty$. Thus at most one "outlier" eigenvalue of X separates from semicircle "bulk":



Heuristic: Suppose $\hat{\lambda} := \lambda_1(X) > 2 + \varepsilon$ is this outlier, w/ eigenvector $\hat{v}$.

Then $\hat{\lambda}\hat{v} = X\hat{v} = \lambda(\hat{v}^T v)v + W\hat{v}$

$$\Rightarrow -\lambda(\hat{v}^T v)v = (W - \hat{\lambda} I)\hat{v}$$

$$\Rightarrow \hat{v} = -\lambda(\hat{v}^T v) R(\hat{\lambda}) v \qquad (R(z) = (W - zI)^{-1})$$

$$\Rightarrow \hat{v}^T v = -\lambda(\hat{v}^T v) \cdot v^T R(\hat{\lambda}) v$$

$$\Rightarrow -\frac{1}{\lambda} = v^T R(\hat{\lambda}) v.$$

Deterministic equivalent for $R(z)$ is $m_{sc}(z) I$, so $v^T R(\hat{\lambda}) v \approx m_{sc}(\hat{\lambda})$

$\Rightarrow -\frac{1}{\lambda} \approx m_{sc}(\hat{\lambda})$. Note $m_{sc}(z) \in (-1, 0)$ for $z \in (2, \infty)$, so this is only possible if $\lambda > 1$. Then

$$0 = m_{sc}(z)^2 + z m_{sc}(z) + 1 \Rightarrow 0 \approx \frac{1}{\lambda^2} - \frac{\hat{\lambda}}{\lambda} + 1$$

$$\Rightarrow \boxed{\hat{\lambda} \approx \lambda + \frac{1}{\lambda}}$$

Thm: Let $X = \sum_{i=1}^{k} \lambda_i v_i v_i^T + W$, where W is Wigner, $v_1, \dots, v_k$ are orthogonal unit vectors, $\lambda_1 \geq \dots \geq \lambda_k \geq 0$ fixed as $n \to \infty$.

(a) Let $\hat{\lambda}_i$ be i^{th} largest eigenvalue of X. Then for each $i \in \{1, \dots, k\}$:

$$\hat{\lambda}_i \to \begin{cases} \lambda_i + \frac{1}{\lambda_i} & \text{if } \lambda_i > 1 \\ 2 & \text{if } \lambda_i \in [0, 1] \end{cases} \quad \text{a.s.}$$

(b) Let $\hat{v}_i$ be unit eigenvector of X for $\hat{\lambda}_i$. If $\lambda_i > 1$ and $\lambda_i \neq \lambda_{i-1}, \lambda_{i+1}$, then

$$(\hat{v}_i^T v_i)^2 \to 1 - \frac{1}{\lambda_i^2} \text{ a.s.}, \quad (\hat{v}_i^T v_j)^2 \to 0 \text{ a.s. } \forall j \in \{1, \dots, k\} \setminus \{i\}$$

Three phenomena:

(1) Phase transition at $\lambda_i < 1$ vs $\lambda_i > 1$. ("BBP phase transition")

(2) Upward bias: $\hat{\lambda}_i > \lambda_i$ for $\lambda_i > 1$.

(3) Eigenvector inconsistency: $(\hat{v}_i^T v_i)^2 < 1$ for $\lambda_i > 1$.

Lemma: Fix $\varepsilon, C > 0$, $\mathbb{U} = \{z \in \mathbb{C} : |z| \le C, \operatorname{Re} z \ge 2 + \varepsilon\}$. Let $R(z) = (W - zI)^{-1}$ where W Wigner. Then for any fixed $i \ne j$:

$$\sup_{z \in \mathbb{U}} |R(z)_{ii} - m_{sc}(z)| \to 0 \text{ a.s.}, \quad \sup_{z \in \mathbb{U}} |R(z)_{ij}| \to 0 \text{ a.s.}$$

Proof: For fixed $z \in \mathbb{U}$, by Schur complement,

$$R(z)_{ii} = \frac{1}{w_{ii} - z - w_i^T R^{(i)}(z) w_i} \to \frac{1}{-z - m_{sc}(z)} = m_{sc}(z) \text{ a.s. (Lecture 3)}$$

$$R(z)_{ij} = -R(z)_{ii} \cdot \underbrace{w_i^T R^{(i)}(z) e_j}_{< \frac{1}{\sqrt{n}} \|R^{(i)}(z) e_j\|_2 < \frac{1}{\sqrt{n}}} \to 0 \text{ a.s. (Lecture 7)}$$

For any countable dense $\mathbb{D} \subset \mathbb{U}$, this implies

$$\sup_{z \in \mathbb{D}} |R(z)_{ii} - m_{sc}(z)| \to 0, \quad \sup_{z \in \mathbb{D}} |R(z)_{ij}| \to 0 \text{ a.s.}$$

On the event where $\|W\|_{op} < 2 + \frac{\varepsilon}{2}$ (holds a.s. all large n), $R(z)_{ii} - m_{sc}(z)$ and $R(z)_{ij}$ analytic on $\mathbb{U} \Rightarrow$ uniform convergence extends to $\mathbb{U}$. ∎

[This is enough to prove Theorem for $W \sim GOE$. For general Wigner W, need $\sup_{z \in \mathbb{U}} |u^T R(z) v - u^T v \cdot m_{sc}(z)| \to 0$ $\forall$ deterministic unit vectors u, v.]

Proof of Thm ($W \sim GOE$): Let $V = [v_1, \dots, v_k] \in \mathbb{R}^{n \times k}$, $\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_k \end{pmatrix}$, $O \in \mathbb{R}^{n \times n}$ any completion of V to an orthonormal basis. Then

$$X = V\Lambda V^T + W$$

$$= O\underbrace{\left([e_1, \dots, e_k]\Lambda[e_1, \dots, e_k]^T + \tilde{W}\right)}_{:=\tilde{X}}O^T \qquad \left(\tilde{W} := O^T W O \overset{d}{=} W \text{ by rotational-invariance of GOE}\right)$$

- Eigenvalues of X are equal to those of $\tilde{X}$.
- Eigenvectors of X are $\hat{v}_i = O\tilde{v}_i$ where $\tilde{v}_i$ is eigenvector of $\tilde{X}$

$$\Rightarrow (\hat{v}_i^T v_j)^2 = (\tilde{v}_i^T O^T v_j)^2 = (\tilde{v}_i^T e_j)^2 \quad \forall i, j \in \{1, \dots, k\}.$$

Hence: Suffices to prove result for $\tilde{X}$, i.e. we may assume WLOG

$$V = [e_1, \dots, e_k].$$

(a) Fix $\varepsilon > 0$. Let $\mathcal{E}_n$ be event where $\|W\|_{op} < 2 + \frac{\varepsilon}{2}$.

On $\mathcal{E}_n$: $\|X\|_{op} < 2 + \max_{i=1,\dots,k} |\lambda_i| + \frac{\varepsilon}{2} := C$.

Eigenvalues $\hat{\lambda} \in (2+\varepsilon, C)$ of X are roots of

$$0 = \det(X - \hat{\lambda} I)$$

$$= \det(W - \hat{\lambda} I + V\Lambda V^T)$$

$$= \det(W - \hat{\lambda} I)\det(I + R(\hat{\lambda})V\Lambda V^T) \qquad (\det AB = \det A \det B)$$

Since $\det(W - \hat{\lambda} I)$ has no roots in $(2+\varepsilon, C)$:

$$0 = \det(I + R(\hat{\lambda})V\Lambda V^T)$$

$$= \det(I + V^T R(\hat{\lambda})V\Lambda) \qquad (\det(I+AB) = \det(I+BA))$$

Let $\mathbb{U} = \{z \in \mathbb{C} : |z| \le C, \ \mathrm{Re}\, z \ge 2 + \varepsilon\}$. Assuming $V = [e_1, \dots, e_k]$, by lemma:

$$\sup_{z \in \mathbb{U}} \left| \det(I + V^T R(z) V \Lambda) - \det(I + m_{sc}(z)\Lambda) \right| \to 0 \quad \text{a.s.} \quad (*)$$

We apply Rouche's Theorem: If f, g are analytic on $K \subseteq \mathbb{C}$ w/

$|g(z)| < |f(z)|$ on ∂K, then $f, f+g$ have same # roots in K.

Take $f(z) = \det(I + m_{sc}(z)\mathcal{N}) = \prod_{i=1}^{k} (1 + m_{sc}(z)\lambda_i)$.

Here: For $z \in \mathbb{C}^+$: $m_{sc}(z) \in \mathbb{C}^+$

$z \in \mathbb{C}^-$: $m_{sc}(z) \in \mathbb{C}^-$

$z \in (2, \infty)$: $m_{sc}(z) = \frac{-z + \sqrt{z^2-4}}{2}$, increasing from -1 to 0.

$$m_{sc}(z) = -\frac{1}{\lambda} \text{ for } \lambda > 1 \iff z = \lambda + \frac{1}{\lambda}.$$

Thus, for any $\varepsilon > 0$ small enough, roots of f in $\mathcal{U}$ are

$$\left\{\lambda_i + \frac{1}{\lambda_i}\right\}_{i=1}^{k_+}, \quad k_+ \text{ largest index s.t. } \lambda_{k_+} > 1.$$

By Rouche's Thm and (*) applied to (1) $\mathcal{U}$ and (2) ball of radius δ and each root of f, taking $\delta \to 0$:

- $\det(I + V^T R(z) V \cdot \mathcal{N})$ also has k_+ roots in $\mathcal{U}$ a.s. all large n.
- These roots converge a.s. to $\left\{\lambda_i + \frac{1}{\lambda_i}\right\}_{i=1}^{k_+}$.

Thus for any $\varepsilon > 0$ small enough, a.s. for all large n, X has k_+ eig. values larger than $2+\varepsilon$, converging to $\left\{\lambda_i + \frac{1}{\lambda_i}\right\}_{i=1}^{k_+}$. Taking $\varepsilon \to 0$ shows (a).

(b) Fix $i \in \{1, \dots, k_+\}$. Then (assuming $V = [e_1, \dots, e_k]$)

$$\hat{\lambda}_i \hat{v}_i = X \hat{v}_i = W \hat{v}_i + \sum_{\ell=1}^{k} \lambda_\ell \hat{v}_i^T e_\ell e_\ell$$

On $\mathcal{E}_n$, this implies $-(W - \hat{\lambda}_i I)\hat{v}_i = \sum_{\ell=1}^{k} \lambda_\ell \hat{v}_i^T e_\ell \cdot e_\ell$

$$\Rightarrow \hat{v}_i = -\sum_{\ell=1}^{k} R(\hat{\lambda}_i) e_\ell \cdot \lambda_\ell \hat{v}_i^T e_\ell \qquad (**)$$

For any $j \neq i$: $\hat{v}_i^T e_j = -\sum_{\ell=1}^{k} R(\hat{\lambda}_i)_{j\ell} \cdot \lambda_\ell \cdot \hat{v}_i^T e_\ell$

Apply
- $R(\hat{\lambda}_i)_{j\ell} \to 0$ a.s. $\forall \ell \neq j \in \{1, \dots, k\}$
- $R(\hat{\lambda}_i)_{jj} - m_{sc}(\hat{\lambda}_i) \to 0$ a.s. $\forall j = 1, \dots, k$
- $m_{sc}(\hat{\lambda}_i) \to m_{sc}(\lambda_i + \frac{1}{\lambda_i}) = -\frac{1}{\lambda_i}$ a.s.

$$\Rightarrow \left| \hat{v}_i^T e_j - \frac{\lambda_j}{\lambda_i} \hat{v}_i^T e_j \right| \to 0 \text{ a.s.}$$

$$\Rightarrow |\hat{v}_i^T e_j| \to 0 \text{ a.s. since } \lambda_j \neq \lambda_i.$$

Then $(**) \Rightarrow \hat{v}_i = -R(\hat{\lambda}_i) e_i \cdot \lambda_i \cdot \hat{v}_i^T e_i + r_n$, $\|r_n\|_2 \to 0$ a.s.

Take squared norm, w/ $\|\hat{v}_i\|_2^2 = 1$: $\lambda_i^2 (\hat{v}_i^T e_i)^2 \cdot [R(\hat{\lambda}_i)^2]_{ii} \to 1$ a.s.

Recall $R(z)^2 = R'(z)$ (Lecture 6). By previous lemma and analyticity, $\sup_{z \in \mathbb{C}} |R'(z)_{ii} - m_{sc}'(z)| \to 0$, so $[R(\hat{\lambda}_i)^2]_{ii} - m_{sc}'(\hat{\lambda}_i) \to 0$ a.s.

Here $m_{sc}(z)^2 + z m_{sc}(z) + 1 = 0 \Rightarrow (2 m_{sc}(z) + z) m_{sc}'(z) + m_{sc}(z) = 0$

$$\Rightarrow m_{sc}'(z) = -\frac{m_{sc}(z)}{2 m_{sc}(z) + z}$$

So $m_{sc}'(\hat{\lambda}_i) \to m_{sc}'(\lambda_i + \frac{1}{\lambda_i}) = -\frac{-1/\lambda_i}{-2/\lambda_i + \lambda_i + 1/\lambda_i} = \frac{1}{\lambda_i^2 - 1}$ a.s.

$$\Rightarrow (\hat{v}_i^T e_i)^2 \to \frac{1}{\lambda_i^2 \cdot \frac{1}{\lambda_i^2 - 1}} = 1 - \frac{1}{\lambda_i^2} \text{ a.s.}$$

Spiked covariance model

Let $x_1, \dots, x_n$ be iid vectors in $\mathbb{R}^d$, $\mathbb{E} x_i = 0$, $\mathbb{E} x_i x_i^T = \Sigma$.

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^{n} x_i x_i^T \quad \text{(sample covariance)}$$

Recall (Lecture 5): If $\Sigma = I$, $\frac{d}{n} \to \gamma \in (0, \infty)$, then

ESD of $\hat{\Sigma} \to \mu_{MP}$ weakly a.s.,

$$d\mu_{MP}(x) = \frac{1}{2\pi\gamma x} \sqrt{(\lambda_+ - x)(x - \lambda_-)} \quad \forall x > 0, \quad \lambda_\pm = (1 \pm \sqrt{\gamma})^2.$$

Suppose eigenvalues of Σ are instead $(\lambda_1, \dots \lambda_k, 1, \dots 1)$.

k is fixed, so limit ESD of Σ is same as for $\Sigma = I$.

What is the behavior of eigenvalues/eigenvectors (i.e. PCs) of $\hat{\Sigma}$?

Thm: Suppose $\Sigma = I$. Then $\lambda_1(\hat{\Sigma}) \to \lambda_+ = (1+\sqrt{\gamma})^2$ a.s.

Proof: Either moment or resolvent method.

[Geman '80, Bai-Silverstein '98]

Thm: Under the same assumptions as Lecture 5, suppose

Σ has eigenvalues $\lambda_1 \geq \dots \geq \lambda_k > 1 = \lambda_{k+1} = \dots = \lambda_d$ (fixed w/ n, d).

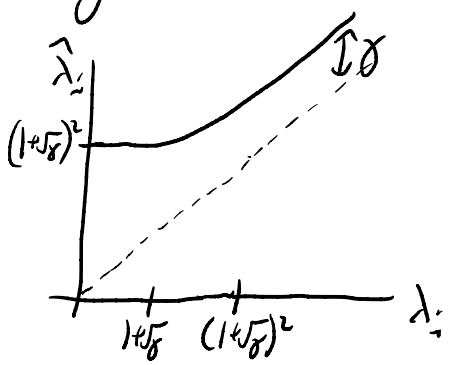
(a) Let $\hat{\lambda}_i$ be i^{th} largest eigenvalue of $\hat{\Sigma}$. Then for each $i = 1, \dots, k$,

$$\hat{\lambda}_i \to \begin{cases} \lambda_i \left(1 + \frac{\gamma}{\lambda_i - 1}\right) & \text{if } \lambda_i > 1 + \sqrt{\gamma} \\ \lambda_+ = (1+\sqrt{\gamma})^2 & \text{if } \lambda_i \in [1, 1+\sqrt{\gamma}] \end{cases} \quad \text{a.s.}$$

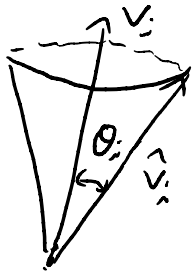
(b) Let $\hat{v}_i, v_i$ be unit eigenvectors of $\hat{\Sigma}, \Sigma$ for $i = 1, \dots, k$. If $\lambda_i > 1 + \sqrt{\gamma}$ and $\lambda_i \neq \lambda_{i-1}, \lambda_{i+1}$, then

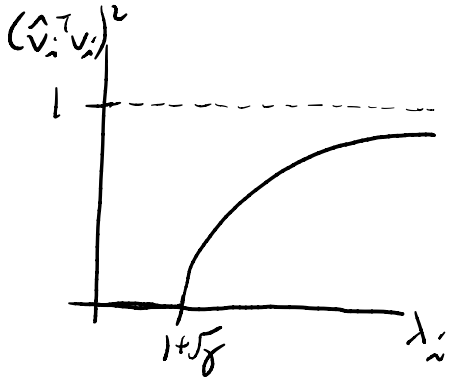
$$(\hat{v}_i^T v_i)^2 \to \frac{1 - \frac{\gamma}{(\lambda_i - 1)^2}}{1 + \frac{\gamma}{\lambda_i - 1}} \text{ a.s.}, \quad (\hat{v}_i^T v_j)^2 \to 0 \ \ \forall j \in \{1, \dots, k\} \setminus \{i\},$$

Eigenvalue bias:



Eigenvector alignment: $|\hat{v}_i^T v_i| = \cos \theta_i$





Proof (Gaussian case, $x_1, \dots, x_n \overset{iid}{\sim} \mathcal{N}(0, \Sigma)$):

Suppose $\Sigma = V \Lambda V^T$, $\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_k \\ & & & 1 \\ & & & & \ddots \end{pmatrix}$.

Then $\hat{\Sigma} = \Sigma^{1/2} Z^T Z \Sigma^{1/2}$ $\quad (Z = (z_{ij})_{i \in [n], j \in [d]}$ w/ $z_{ij} \overset{iid}{\sim} \mathcal{N}(0, \frac{1}{n}))$

$$= V \underbrace{\left(\Lambda^{1/2} \tilde{Z}^T \tilde{Z} \Lambda^{1/2} \right)}_{:= \tilde{\Sigma}} V^T \quad (\tilde{Z} = ZV \overset{d}{=} Z)$$

- Eigenvalues of $\hat{\Sigma}$ are equal to those of $\tilde{\Sigma}$
- Eigenvectors of $\hat{\Sigma}$ are $\hat{v}_i = V \tilde{v}_i$

$$\Rightarrow (\hat{v}_i^T v_j)^2 = (\tilde{v}_i^T V^T v_j)^2 = (\tilde{v}_i^T e_j)^2 \quad \forall i, j \in \{1, \dots, k\}.$$

Hence: Assume WLOG $\Sigma = \Lambda$, w/ eigenvectors $e_1, \dots, e_d$. Let

$$\Lambda = \begin{pmatrix} \Lambda_1 & 0 \\ 0 & I \end{pmatrix} \text{ where } \Lambda_1 = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_k \end{pmatrix}$$

$$\hat{\Sigma} = \begin{pmatrix} \Lambda_1^{1/2} & 0 \\ 0 & I \end{pmatrix} \underbrace{\begin{pmatrix} Z_1^T Z_1 & Z_1^T Z_2 \\ Z_2^T Z_1 & Z_2^T Z_2 \end{pmatrix}} \begin{pmatrix} \Lambda_1^{1/2} & 0 \\ 0 & I \end{pmatrix} := \begin{pmatrix} \hat{\Sigma}_{11} & \hat{\Sigma}_{12} \\ \hat{\Sigma}_{21} & \hat{\Sigma}_{22} \end{pmatrix}$$

$$= Z^T Z \text{ where } Z = (\underbrace{Z_1}_{k} \; \underbrace{Z_2}_{d-k})$$

Fix $\varepsilon > 0$, let $\mathcal{E}_n$ be event $\|Z_2^T Z_2\|_{op}, \|Z^T Z\|_{op} < (1+\sqrt{\gamma})^2 + \frac{\varepsilon}{2}$.

(a) On $\mathcal{E}_n$: $\|\hat{\Sigma}\|_{op} < \left[(1+\sqrt{\gamma})^2 + \frac{\varepsilon}{2}\right] \cdot \max_{i=1}^{k} \lambda_i := C$.

Eigenvalues $\hat{\lambda} \in \left((1+\sqrt{\gamma})^2 + \varepsilon, C\right)$ of $\hat{\Sigma}$ are roots of

$$0 = \det(\hat{\Sigma} - \hat{\lambda} I)$$

$$= \det\left(\hat{\Sigma}_{22} - \hat{\lambda} I\right) \cdot \det\left(\hat{\Sigma}_{11} - \hat{\lambda} I - \hat{\Sigma}_{12} \underbrace{\left(\hat{\Sigma}_{22} - \hat{\lambda} I\right)^{-1}}_{:= R(\hat{\lambda})} \hat{\Sigma}_{21}\right)$$

Since $\det(\hat{\Sigma}_{22} - \hat{\lambda} I)$ has no roots in $\left((1+\sqrt{\gamma})^2 + \varepsilon, C\right)$:

$$0 = \det\left(\hat{\Sigma}_{11} - \hat{\lambda} I - \hat{\Sigma}_{12} R(\hat{\lambda}) \hat{\Sigma}_{21}\right)$$

$$= \det\left(\Lambda_1^{1/2} Z_1^T Z_1 \Lambda_1^{1/2} - \hat{\lambda} I - \Lambda_1^{1/2} Z_1^T Z_2 R(\hat{\lambda}) Z_2^T Z_1 \Lambda_1^{1/2}\right)$$

$$= \det\left(\Lambda_1^{1/2} Z_1^T \left(I - Z_2 R(\hat{\lambda}) Z_2^T\right) Z_1 \Lambda_1^{1/2} - \hat{\lambda} I\right)$$

Let $\mathbb{U} = \{z \in \mathbb{C} : |z| \le C, \operatorname{Re} z \ge (1+\sqrt{\gamma})^2 + \varepsilon\}$,

$Z_1 = [z_1, \dots, z_k]$ (independent of Z_2, $R(z)$).

For any $z \in \mathcal{U}$, $i \neq j \in \{1, \dots, k\}$,

- $$\left| z_i^T (I - Z_2 R(z) Z_2^T) z_j - \frac{1}{n} \mathrm{Tr}\,(I - Z_2 R(z) Z_2^T) \right|$$

$$\prec \left\| \frac{1}{n} (I - Z_2 R(z) Z_2^T) \right\|_F$$

$$\left| z_i^T (I - Z_2 R(z) Z_2^T) z_j \right| \prec \left\| \frac{1}{n} (I - Z_2 R(z) Z_2^T) \right\|_F$$

- $$I - Z_2 R(z) Z_2^T = I - Z_2 (Z_2^T Z_2 - zI)^{-1} Z_2^T$$

$$= \left(I - \frac{1}{z} Z_2 Z_2^T\right)^{-1} \quad \text{(Woodbury identity)}$$

$$= -z (Z_2 Z_2^T - zI)^{-1}$$

$$\Rightarrow \left\| I - Z_2 R(z) Z_2^T \right\|_{op} \leq \frac{C}{\varepsilon/2} \quad \text{on } \mathcal{C}_n.$$

$$\frac{1}{n} \mathrm{Tr}\,(I - Z_2 R(z) Z_2^T) \to -z \underline{m}(z) \quad \text{a.s.}$$

Then by analyticity:

$$\left.\begin{aligned} &\sup_{z \in \mathcal{U}} \left| z_i^T (I - Z_2 R(z) Z_2^T) z_i - (-z \underline{m}(z)) \right| \to 0 \quad \text{a.s.} \quad \forall i = 1, \dots, k \\ &\sup_{z \in \mathcal{U}} \left| z_i^T (I - Z_2 R(z) Z_2^T) z_j \right| \to 0 \quad \text{a.s.} \quad \forall i \neq j \in \{1, \dots, k\} \end{aligned}\right\} (*)$$

Then

$$\sup_{z \in \mathcal{U}} \left| \det\left(\hat{\Sigma}_{11} - zI - \hat{\Sigma}_{12} R(z) \hat{\Sigma}_{21}\right) - \det\left(-z(\underline{m}(z) \Lambda_1 + I)\right) \right| \to 0 \quad \text{a.s.}$$

Here: $\det\left(-z(\underline{m}(z)\Lambda_1 + I)\right) = (-z)^k \prod_{i=1}^{k} (\underline{m}(z)\lambda_i + 1)$

$\underline{m}(z)$ solves $z = -\frac{1}{\underline{m}(z)} + \gamma \frac{1}{1+\underline{m}(z)}$

$$\Rightarrow \underline{m}(z) = \frac{\gamma - 1 - z + \sqrt{(z-\lambda_-)(z-\lambda_+)}}{2z} \quad \text{for } z > \lambda_+ = (1+\sqrt{\gamma})^2$$

$\underline{m}(z)$ increases from $\frac{\gamma - 1 - \lambda_+}{2\lambda_+} = -\frac{1}{1+\sqrt{\gamma}}$ to 0 over $z \in (\lambda_+, \infty)$.

For $\lambda > 1+\sqrt{\gamma}$, $\underline{m}(z) = -\frac{1}{\lambda} \iff z = \lambda + \frac{\gamma}{1-1/\lambda} = \lambda\left(1 + \frac{\gamma}{\lambda - 1}\right)$.

So by Rouche's Thm, for any $\varepsilon > 0$ sufficiently small, roots of $0 = \det(\hat{\Sigma}_{11} - zI - \hat{\Sigma}_{12} R(z) \hat{\Sigma}_{21})$ in $\mathbb{U}$ converge to

$$\left\{ \lambda_i \left(1 + \frac{\gamma}{\lambda_i - 1}\right) \right\}_{i=1}^{k_+}, \quad k_+ \text{ largest index s.t. } \lambda_{k_+} > (1+\sqrt{\gamma})^2.$$

Taking $\varepsilon \to 0$ shows (a).

(b) Let $\hat{v}_i = (\hat{v}_{i1}, \hat{v}_{i2})$. Then

$$\hat{\lambda}_i \hat{v}_i = \hat{\Sigma} \hat{v}_i \Rightarrow \hat{\lambda}_i \begin{pmatrix} \hat{v}_{i1} \\ \hat{v}_{i2} \end{pmatrix} = \begin{pmatrix} \hat{\Sigma}_{11} & \hat{\Sigma}_{12} \\ \hat{\Sigma}_{21} & \hat{\Sigma}_{22} \end{pmatrix} \begin{pmatrix} \hat{v}_{i1} \\ \hat{v}_{i2} \end{pmatrix}$$

$$\Rightarrow \begin{cases} 0 = (\hat{\Sigma}_{11} - \hat{\lambda}_i I) \hat{v}_{i1} + \hat{\Sigma}_{12} \hat{v}_{i2} \\ 0 = (\hat{\Sigma}_{22} - \hat{\lambda}_i I) \hat{v}_{i2} + \hat{\Sigma}_{21} \hat{v}_{i1} \end{cases}$$

Second equation shows $\hat{v}_{i2} = -R(\hat{\lambda}_i) \hat{\Sigma}_{21} \hat{v}_{i1}$

$$\Rightarrow \begin{cases} 0 = (\hat{\Sigma}_{11} - \hat{\lambda}_i I - \hat{\Sigma}_{12} R(\hat{\lambda}_i) \hat{\Sigma}_{21}) \hat{v}_{i1} & (**) \\ 1 = \|\hat{v}_{i1}\|_2^2 + \|\hat{v}_{i2}\|_2^2 = \hat{v}_{i1}^\top (I + \hat{\Sigma}_{12} R(\hat{\lambda}_i)^2 \hat{\Sigma}_{21}) \hat{v}_{i1} & (***) \end{cases}$$

By (*) and part (a):

$$\hat{\Sigma}_n - \hat{\lambda}_i I - \hat{\Sigma}_{12} R(\hat{\lambda}_i) \hat{\Sigma}_{21}$$

$$\to -\lambda_i \left(1 + \frac{\gamma}{\lambda_i - 1}\right)\Big(\underbrace{\underline{m}\left(\lambda_i\left(1+\frac{\gamma}{\lambda_i-1}\right)\right)}_{= -\frac{1}{\lambda_i}} \Lambda_1 + I \Big) \quad \text{a.s.}$$

Since $\lambda_i \neq \lambda_{i-1}, \lambda_{i+1}$, this is diagonal w/ j^{th} diagonal entry non-zero $\forall j \neq i$. So (**) implies $\hat{v}_i^T e_j = \hat{v}_{i1}^T e_j \to 0$ a.s. $\forall j \neq i$.

Then $|\hat{v}_i^T e_i| = \|\hat{v}_{i1}\|_2 + o(1)$ a.s., so (***) implies

$$(\hat{v}_i^T e_i)^2 \cdot \left(I + \hat{\Sigma}_{12} R(\hat{\lambda}_i)^2 \hat{\Sigma}_{21}\right)_{ii} \to 1 \quad \text{a.s.}$$

Here $\left(I + \hat{\Sigma}_{12} R(z)^2 \hat{\Sigma}_{21}\right)_{ii} = -\frac{d}{dz}\left(\hat{\Sigma}_{11} - zI - \hat{\Sigma}_{12} R(z) \hat{\Sigma}_{21}\right)_{ii}$

$$\to \frac{d}{dz}\left(z\left(\underline{m}(z)\Lambda_1 + I\right)\right)_{ii} \quad \text{a.s.}$$

$$= \underline{m}(z)\lambda_i + 1 + z\underline{m}'(z)\lambda_i$$

$$\Rightarrow (\hat{v}_i^T e_i)^2 \to \frac{1}{\lambda_i\left(1+\frac{\gamma}{\lambda_i-1}\right) \cdot m'\left(\lambda_i\left(1+\frac{\gamma}{\lambda_i-1}\right)\right) \cdot \lambda_i} \quad \text{a.s.}$$

$$z = -\frac{1}{m(z)} + \frac{\gamma}{1+m(z)} \Rightarrow m'(z) = \frac{1}{\frac{1}{m(z)^2} - \frac{\gamma}{(1+m(z))^2}}$$

$$\Rightarrow m'\left(\lambda_i\left(1+\frac{\gamma}{\lambda_i-1}\right)\right) = \frac{1}{\lambda_i^2 - \frac{\gamma}{(1-1/\lambda_i)^2}} = \frac{1}{\lambda_i^2\left(1-\frac{\gamma}{(\lambda_i-1)^2}\right)}$$

$$\Rightarrow (\hat{v}_i^T e_i)^2 \to \frac{1-\frac{\gamma}{(\lambda_i-1)^2}}{1+\frac{\gamma}{\lambda_i-1}} \quad \text{a.s.}$$

Problem 8: Assume Thm holds for $X = \lambda vv^T + W$ when W is any Wigner matrix, and $\sup_{z \in \Omega} |u^T (W - zI)^{-1} v - u^T v \cdot m_{sc}(z)| < \frac{1}{\sqrt{n}}$ for any deterministic unit vectors u, v.

(a) Consider stochastic block model: $\{1, \dots, n\} = S_1 \cup S_2$, $|S_1| = |S_2| = \frac{n}{2}$.

Independently for each vertex pair $i \neq j \in \{1, \dots, n\}$:

$$\mathbb{P}[i \sim j] = \begin{cases} \frac{1}{2} + \frac{c}{\sqrt{n}} & \text{if both } i, j \in S_1 \text{ or both } i, j \in S_2 \\ \frac{1}{2} - \frac{c}{\sqrt{n}} & \text{otherwise} \end{cases}$$

Let $a_{ij} = a_{ji} = \frac{\mathbb{1}\{i \sim j\} - 1/2}{\sqrt{n/4}}$, $a_{ii} = 0$, $A = (a_{ij})_{i,j=1}^n$. As $n \to \infty$, for which values $c > 0$ does the leading eigenvector $\hat{v}$ of A form a non-vanishing angle w/ $u = (u_i)_{i=1}^n$, $u_i = \begin{cases} +1 & i \in S_1 \\ -1 & i \in S_2 \end{cases}$?

(b) Show that $\hat{v}$ is proportional to $\tilde{v} = R(\hat{\lambda}) u$, where $\hat{\lambda}$ is the eigenvalue of A and $R(z)$ is the resolvent of a (nearby) Wigner matrix W. For any entry $i \in \{1, \dots, n\}$, applying

$$\tilde{v}_i = R(\hat{\lambda})_{ii} u_i + \sum_{j \neq i} R(\hat{\lambda})_{ij} u_j = R(\hat{\lambda})_{ii} \left[u_i - w_i^T R^{(i)}(\hat{\lambda}) u^{(i)} \right]$$

show that $\tilde{v}_i$ converges in distribution as $n \to \infty$.

(c) Consider the estimator $\hat{u}$ w/ entries $\hat{u}_i = \operatorname{sign}(\hat{v}_i)$.

Derive an expression for $\lim_{n \to \infty} \min_{s \in \{+1, -1\}} \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n \mathbb{1}\{s \hat{u}_i \neq u_i\} \right]$.